Fiscal Year:	FY 2021	**Task Last Updated:**	FY 05/28/2021
PI Name:	Miller, Fletcher J Ph.D.		
Project Title:	Developing the Narrow Channel Apparatus as a NASA Standard Material Flammability Test		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	fletcher.miller@sdsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	619-594-5791
Organization Name:	San Diego State University		
PI Address 1:	5500 Campanile Drive		
PI Address 2:	Department of Mechanical Engineering		
PI Web Page:			
City:	San Diego	**State:**	CA
Zip Code:	92182-1323	**Congressional District:**	53
Comments:	Previously at National Center for Space Exploration Research, NASA Glenn Research Center, until 2007.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	08/16/2019	**End Date:**	08/15/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
Contact Email:	Sandra.Olson@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Olson, Sandra Ph.D. (NASA Glenn Research Center) Wichman, Indrek Ph.D. (Michigan State University/ grant 80NSSC21M0131)		
Grant/Contract No.:	80NSSC19M0212		
Performance Goal No.:			
Performance Goal Text:			
	NOTE (Ed., November 2019): Continuation of "Developing the Narrow Channel Apparatus as a NASA Standard Material Flammability Test," grant #NNX10AD96A, 1/1/2010 to 12/31/2014; and "Developing the Narrow Channel Apparatus as a NASA Standard Material Flammability Test (NNX14AC70A)," 2/1/2014 to 1/31/2017, with the same Principal Investigator Dr. Fletcher Miller. The research proposed here is directed toward further development of a new apparatus to replace or augment the current NASA Standard 6001 Test 1 used to screen materials for space flight applications, and specifically the flight experiment that will allow us to obtain flame spread data on thick fuel. Test 1 is by far the most widely used test in the Standard, but does not take into account the influence of flow in a controlled manner, while flow has been shown to have a large (often dominant) effect on flame spread in microgravity. The ground-based apparatus described here, The Narrow		

Task Description:	Channel Apparatus, has been used by the author and collaborators for several years to study fundamental aspects of material flammability in reduced buoyancy conditions. Two new versions of the apparatus have been built that allow the study of flame spread on thick fuels under low pressure, high oxygen conditions such as being considered for new space craft, as well as allow examination of effects of apparatus orientation and sample width. Replacing the Test 1 apparatus with a Narrow Channel will allow for rapid, efficient testing, while also taking into account the low flows actually encountered in microgravity. By controlling the flow, it should be possible to approximate partial gravity conditions; although it is not a major part of this work, the possibility will be explored in the new vertical narrow channel. This phase of the research continues to be devoted to ground-based testing while also preparing for the flight experiments. The work is conducted collaboratively at three institutions by investigators that have worked together for several years in this area (Dr. S. Olson at NASA Glenn Research Center and Dr. I. Wichman at Michigan State University are the other two investigators). The research at San Diego State University consists of five main tasks. The first task will be to make flame spread measurements in a Narrow Channel in low pressure, high oxygen (normoxic) conditions on thick Poly(methyl methacrylate) (PMMA) to match the planned flight test matrix. The second task involves extending our existing Fire Dynamics Simulator (FDS) and/or FLUENT model of flame spread in a Narrow Channel to evaluate the influence of heat loss to the upper wall as well as determine velocity and other field data not easily accessible experimentally. In this part of the research FDS will be run with Gpyro to determine the effect of solid-phase heat conduction. The model will also be run to predict differences between normal, partial, and microgravity including some runs with the fuel surface aligned with gravity. The third task will be to continue to work with NASA to develop and fly a flight experiment to test flame spread with forced flow over a thermally thick fuel in microgravity. In support of this effort, we may do some drop tower tests. There is a large data gap in this subject area, and true microgravity verifications tests are needed to prove the validity of the Narrow Channel Apparatus. After the flight, the data will be analyzed and compared to the model. The fourth task will be to perform Narrow Channel experiments on fuels/solids other than PMMA such as Nomex, Kapton, etc. in a higher oxygen, lower pressure environment. We have a selection of fuels based on discussions with NASA Kennedy Space Center (KSC) and NASA Johnson Space Center (JSC). The last task was added new in 2020 and that was the development of a hybrid Test 1 that is conducted in a vertical orientation in a chamber, but with a low upward flow of the test atmosphere to purge combustion products and allow for a much small apparatus. This will provide comparison data to the Narrow Channel results as well as to tests run previously by NASA in the standard Test 1 apparatus.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research encompasses flame spread and material flammability in narrow channels or gaps. We use this geometry to simulate microgravity conditions. However, many fire situations involve flames in narrow gaps. TWA Flight 800, which crashed off the coast of Canada, was brought down by a small fire spreading through some narrow gaps in the walls and ceiling near the cockpit, for instance. Many building fires involve fire spread inside walls. Fires in computers likewise can occur between the printed circuit boards. Thus, information gained from experiments about how flames spread in various atmospheric conditions in narrow gaps is useful in areas well beyond spacecraft fire safety. The numerical model we are developing will be validated against our experiments and can also be used to study flame spread in these gaps.
Task Progress:	This summary covers the period from August 2019 through August 2021. During this period there were three primary accomplishments. The first accomplishment was the continued development of a technique to use a commercial infrared camera to monitor the surface temperature of the combustible material while it was burning. A Master's student and an undergraduate student worked on this together. They made careful measurements with both the camera and a thermocouple embedded in the surface that accurately measures temperature at a point to calibrate video images acquired by the camera from which the entire time-temperature history of the burning material could be determined. This allows us to understand the extent and mechanism of preheating of the material by the flame as might occur in microgravity. From these measurements we can predict the flame spread rate and use that in models that assess the danger of fire on board spacecraft. This activity was cut short in mid-Spring of 2020 due to the COVID-19 pandemic when we had to stop on-site work in the laboratory. The second accomplishment was the initiation of a computational model of a flame on a surface burning in a low-speed airflow, similar to the ventilation flows on spacecraft. We had previously been using the Fire Dynamics Simulator (FDS) software written by the National Institute for Standards and Technology for this, but in this period we made the switch to FLUENT, a commercial software code. This package is general-purpose heat transfer and fluid flow software. FLUENT has not to our knowledge been used previously for simulating flames on solid materials, but it offers increased execution speed and can better simulate plastics that burn away leaving gaps or valleys in the burning surface. We first created the geometry of our Narrow Channel experimental apparatus in the software and then tried cases of fuel/air mixing without combustion and finally some with combustion and successfully achieved a simulated flame stabilized above the surface. To represent the "valley" created when material burns away, we used experimental data from our partner at Michigan State University that measured the depth of the valley as a function of air velocity. Much of the work in this period has been devoted to obtaining the proper material and combustion properties for the simulations, and running test cases to be sure the results are correct and consistent. This activity has been supported by one graduate student since 2019, and joined by a second one starting in the spring of 2021. Since we can conduct these simulations remotely, this work has not been appreciably affected by COVID-19. The third activity has been the design and construction of a variation of the standard NASA Test 1 apparatus. This was conducted by a team of five undergraduates who took this on for their Senior Design project for the calendar year 2020. We wanted a NASA Test 1 apparatus for our own use to provide comparison data to what we obtain from our own Narrow Channel Apparatus, which we believe is a more representative test for microgravity conditions. The standard NASA test involves a large pressure/vacuum tank that we did not have access to, so we designed a system that uses a much smaller tank and provides a low, make-up air flow that brings fresh oxygen to the flame while sweeping away combustion products. The test is still run in the vertical configuration like the NASA test, and it will be useful to compare results from this new, low-flow test to the NASA results that rely on buoyancy entirely. This may be a more efficient way to run the test since it avoids the long period needed to evacuate the refill in the large chamber. The new apparatus was mostly completed by the end of 2020, but due to COVID-19 restrictions we have not been able to test it yet in the laboratory.

Bibliography Type:	Description: (Last Updated: 06/15/2021)
Abstracts for Journals and Proceedings	Spang P, Miller FJ. "Flammability of Clothing and Food Packaging used in Spacecraft at Normoxic Conditions." Presented at the 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Papers from Meeting Proceedings	Berry M, Miller F, Olson S, Wichman I. "Infrared Measurements of Forward Heat Conduction during Simulated Microgravity Flame Spread in the Narrow Channel Apparatus." 2019 Fall Technical Meeting of the Western States Section of the Combustion Institute, Albuquerque, NM, October 14-15, 2019. 2019 Fall Technical Meeting of the Western States Section of the Combustion Institute, Albuquerque, NM, October 14-15, 2019. https://ntrs.nasa.gov/api/citations/20190033433/downloads/20190033433.pdf ; accessed 6/15/21. , Oct-2019